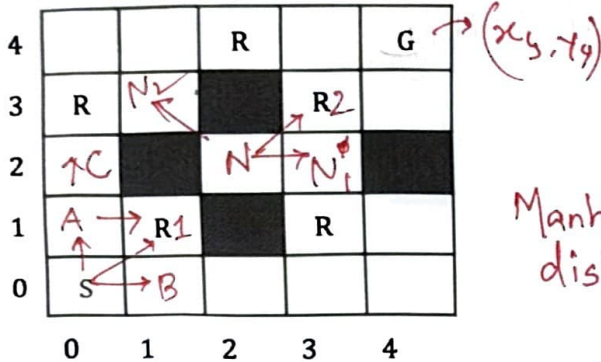


Name:		ID:		Section:	
-------	--	-----	--	----------	--

1.



$$\text{Manhattan distance} = |x_g - x_n| + |y_g - y_n|$$

The given 5X5 table shows a routing network, where a data packet needs to travel from **source S(0, 0)** to **destination G(4, 4)**. Each blank cell represents a **switch** and the cells **marked with R** represent **routers**. The cost of travelling through a switch is 1, but the cost of travelling through a router is 3. The cells marked with **grey colors** represent dead circuits, **through which a data packet cannot pass**.

Movement: The data packet can move UP, RIGHT or diagonally UP.

- Considering **Manhattan Distance** as a heuristic for this problem, simulate A* search on this routing network for TWO iterations. You need to show g, h and f values for each node. Which **three** nodes were **visited** in your simulation? [5]
- Consider the node **N(2, 2)**. Show whether Manhattan Distance is **consistent** for this node. [4]
- Suggest one other heuristic for this problem. Then, show whether your heuristic is **admissible** or not for the node **N(2, 2)**. [4]

(a) Node (N)	g(N)	h(N)	f(N)	parent(N)	PQ: (open list) S A B R1 C
S(0,0)	0	8	8	X	
A(1,0)	1	7	8	S	visited: S A
B(0,1)	1	7	8	S	(closed list)
R1(1,1)	3	6	9	S	
C(2,0)	2	6	8	A	

visited: S, A. Next will be B or C.

(b) $N(2,2) \rightarrow$ Neighbors are $R2(3,3)$ and $N_1(2,3)$ and $N_2(3,1)$

for $N \rightarrow R2$:
 $h(N) = 4, h(R2) = 2, c(N, R2) = 3$
 $4 < 2 + 3 \checkmark$

for $N \rightarrow N_1$:
 $h(N) = 4, h(N_1) = 3$
 $cost(N, N_1) = 1$
 $4 = 3 + 1 \checkmark$

for $N \rightarrow N_2$:
 $h(N) = 4, h(N_2) = 4, c(N, N_2) = 1$
 $4 < 4 + 1 \checkmark$

$\therefore h$ consistent for node N.

(c) Euclidean distance b/w $N(x_N, y_N)$ and $G(x_G, y_G) = \sqrt{(x_G - x_N)^2 + (y_G - y_N)^2}$

$$h(N) = \sqrt{(4-2)^2 + (4-2)^2} = \sqrt{8} = 2.828$$

$h^*(N)$: optimal path from N to G is : $N \rightarrow N_1 \rightarrow N_3 \rightarrow G$

$$\therefore h^*(N) = 3.$$

$h(N) < h^*(N) \rightarrow$ ~~optimal~~ admissible.

2. State whether the following statements are TRUE or FALSE, with a 1-line explanation. [2]

a. If h_1 and h_2 are inadmissible heuristics, $h_3(n) = \min[h_1(n), h_2(n)]$ will be an admissible heuristic.

False. Minimum of h_1 and h_2 can be greater than $h^*(N)$.

$$\text{Let } h_1(N) = 10, h_2(N) = 12, h^*(N) = 8$$

$$h_3(N) = 10 > h^*(N) \rightarrow \text{NOT admissible.}$$

b. An admissible heuristic guarantees that A^* search will be optimal.

False. If the state space is a graph, admissibility is not enough for optimality.

Name:		ID:		Section:	
-------	--	-----	--	----------	--

1.

5					G	(5,5)
4			R		R	
3	R					
2				R ₂		R
1	A					
0	S					
	0	1	2	3	4	5

The given 6X6 table shows a routing network, where a data packet needs to travel from **source S(0, 0)** to **destination G(5, 5)**. Each blank cell represents a **switch** and the cells marked with **R** represent **routers**. The cost of travelling through a switch is 1, but the cost of travelling through a router is 3. The cells marked with **grey colors** represent dead circuits, through which a data packet cannot pass.

Movement: The data packet can move UP, RIGHT or diagonally UP.

- Considering **Manhattan Distance** as a heuristic for this problem, simulate A* search on this routing network for TWO iterations. You need to show g, h and f values for each node. Which **three** nodes were **visited** in your simulation? [5]
- Consider the node **N(2, 2)**. Show whether Manhattan Distance is **consistent** for this node. [4]
- Suggest one other heuristic for this problem. Then, show whether your heuristic is **admissible** or not for the node **N(2, 2)**. [4]

(a) Node (N)	g(N)	h(N)	f(N)	parent(N)	PQ: S A B R ₁ C
S(0,0)	0	10	10	X	visited: S A
A(1,0)	1	9	10	S	
B(0,1)	1	9	10	S	Next node to visit is
R ₁ (1,1)	3	8	11	S	B or C, so first three
C(2,0)	2	8	10	A	will be: S, A, (B or C)

(b) N(2,2) → Neighbors are R₂(2,3), N₁(3,3), N₂(3,1)

for N → R₂:

$$h(N) = 6, h(R_2) = 5, c(N, R_2) = 3$$

$$6 < 5 + 3 \quad \checkmark$$

for N → N₁:

$$h(N) = 6, h(N_1) = 4$$

$$c(N, N_1) = 1$$

$$6 > 4 + 1 \quad \times$$

for N → N₂:

$$h(N) = 6, h(N_2) = 6$$

$$c(N, N_2) = 1$$

$$6 < 6 + 1 \quad \checkmark$$

h is NOT consistent

(c) Another heuristic can be Euclidean Distance.

$$h(N) = \sqrt{(5-2)^2 + (5-2)^2} = \sqrt{9+9} = \sqrt{18} = 4.24$$

$h^*(N)$: shortest path from N to G: $N \rightarrow N1 \rightarrow R \rightarrow G$

$$\therefore h^*(N) = 5$$

$h(N) < h^*(N) \rightarrow$ ~~optimal~~ admissible for node N.

2. State whether the following statements are TRUE or FALSE, with a 1-line explanation. [2]

a. If h_1 and h_2 are admissible heuristics, $h_3(n) = \max[h_1(n), h_2(n)]$ will be inadmissible.

FALSE. can be admissible as well.

$$\text{let } h_1(N) = 5, h_2(N) = 6, h^*(N) = 7$$

$$h_3(N) = 6 < h^*(N) \rightarrow \text{admissible.}$$

b. A consistent heuristic guarantees that A* search will be optimal.

TRUE.